

Regio- and Diastereochemical Aspects of the Additions of Li or Zn Derivatives of Methoxypropene to Oxazolidines Derived from Phenylglycinol

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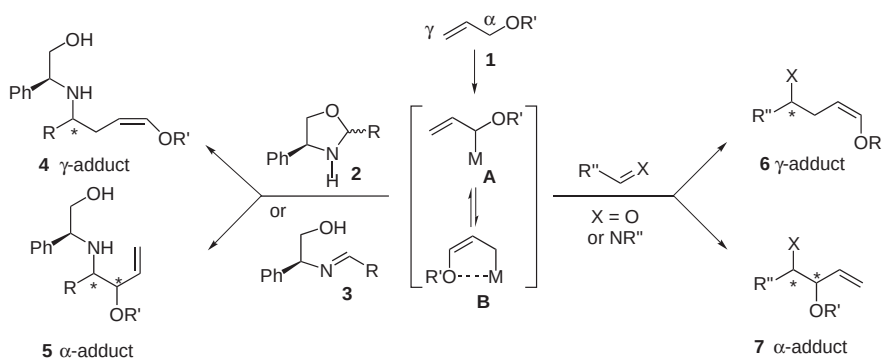
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Abstract: Diastereoselective additions of methoxypropene-derived lithium and zinc reagents to oxazolidines are investigated. The lithium allylic carbanion reacts with oxazolidines to afford mainly the β -amino alcohols with an enol ether function (γ -adduct) and the zinc derivative leads to amino alcohols with an allyl ether function (α -adduct).

Key words: oxazolidine, methoxypropene, organolithium and organozinc reagents, β -amino alcohols

The addition of metallated 1-alkoxy-2-propenes **1** to prochiral electrophiles such as aldehydes or imines is an attractive way for the construction of highly valuable intermediates in organic synthesis. Indeed, depending on the regioselectivity of their addition, they can give access either to the corresponding enol ethers **6** (γ -adducts) or allyl ethers **7** (α -adducts) with the formation of one or two stereogenic centers.¹ If the addition of this type of organometallic reagent to aldehydes has received a considerable attention,² little work on the related additions of such substituted reagents to imines has been reported in the literature.^{3–5} Similarly, if the addition of allylic organometallic reagents to imines^{6,7} or oxazolidines derived from phenylglycinol^{8,4b,4c} is well documented, there is no report, to our knowledge, on the use of α -alkoxyallyl organometallics on these compounds. Such reactions could lead to synthetically useful amino alcohols **4** or **5** (Scheme 1).

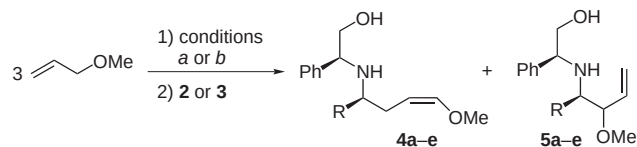


Scheme 1 Reactivity of metallated allylethers with electrophiles

As part of a program aimed at emphasizing the efficiency of asymmetric syntheses of enantiopure cyclic amino acids⁹ and azaheterocycles¹⁰ from new β -amino alcohols, we decided to investigate this topic.

The starting oxazolidines **2** ($R = n\text{-Pr}$, $i\text{-Pr}$, $t\text{-Bu}$) or imines **3** ($R = \text{Ph}$, PhCH=CH) were obtained quantitatively by mixing (*S*)-phenylglycinol with the corresponding aldehydes at room temperature in the presence of MgSO_4 . The reactions were monitored by NMR and the crude materials were used, after filtration and evaporation. As already reported, unsaturated aldehydes yielded the corresponding imines **3**. The imine resulting from the use of cinnamaldehyde was found to be crystalline and therefore was purified in this manner.

The methyl allyl ether **1** ($R' = \text{Me}$) was easily metallated at -78°C by $s\text{-BuLi}$ in THF. Subsequent addition of the oxazolidines **2** or imines **3** to this organolithium derivative afforded a mixture of β -amino alcohols **4a–e** (major isomer) and **5a–e** (minor isomer). These reactions occurred with high regio- and diastereoselectivity (see Scheme 2, conditions *a*, and Table 1).¹¹



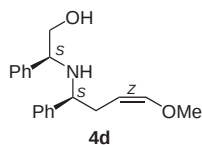
Scheme 2 Reagents and conditions: (*a*) $s\text{-BuLi}$ (3 equiv), THF, -78°C ; (*b*) $s\text{-BuLi}$ (3 equiv), ZnBr_2 (3 equiv), THF, -78°C to r.t.

Table 1 Reactions of the Methyl Allyl Ether Lithium Derivative with Compounds **2** and **3**

Entry	R	4a–e	Yield (%) ^a	4:5 ^b	dr of 4 ^b
1	<i>n</i> -Pr	4a	40	84:16	>95:5
2	<i>i</i> -Pr	4b	62	91:9	>95:5
3	<i>t</i> -Bu	4c	55	93:7	>95:5
4	Ph	4d	67	91:9	91:9
5	Cinnamyl	4e	40	95:5	>95:5

^a Isolated yield of **4a–e**.^b Determined by ¹H NMR of the crude product.

As shown in the Table 1, in all cases, the major regioisomer was the γ -adduct **4** obtained with an acceptable isolated yield. Curiously, the poorest regioselectivity was observed with the linear substituent *n*-Pr (entry 1). The diastereomeric purity of the γ -adducts, measured by ¹H NMR analysis of the crude product, was found to be good to excellent. The configuration of the new stereogenic center and the stereochemistry of the double bond was determined from compound **4d** (R = Ph)¹² by X-ray analysis and found to be respectively *S* and *Z* (Figure 1).¹³ We assume the same configuration for all compounds **4**.¹⁴

**Figure 1**

In a second set of experiments the above reactions were performed using the zinc reagent, prepared by transmetalation of the corresponding lithium derivative by ZnBr₂ (Scheme 2, conditions b). The results obtained are reported in Table 2.¹⁵

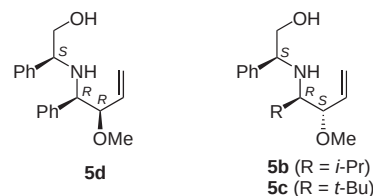
Table 2 Reactions of the Methyl Allyl Ether Zinc Derivative with Compounds **2** and **3**

Entry	R	5a–e	Yield (%) ^a	4:5 ^b	dr of 5 ^b
1	<i>n</i> -Pr	5a	40	>3:97	70:30
2	<i>i</i> -Pr	5b	58	4:96	95:5
3	<i>t</i> -Bu	5c	70	>3:97	>95:5
4	Ph	5d	69	>3:97	96:4

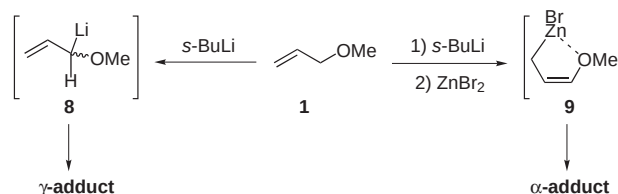
^a Isolated yield of **5a–d**.^b Determined by ¹H NMR of the crude product.

Under these conditions, compounds **5a–d**, resulting from an attack of the organozinc reagent at the α -position, were obtained with moderate to good yields. In three cases, the reaction was totally regioselective and occurred with a satisfactory diastereoselectivity, except in the case of

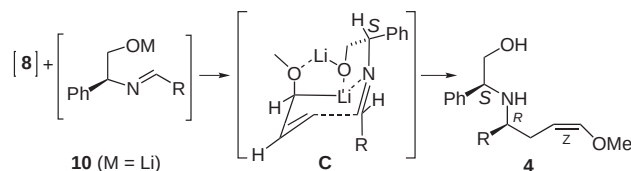
product **5a** (R = *n*-Pr). The absolute configuration of the two new stereogenic centers was determined by X-ray analyses¹⁶ of compound **5d**¹⁷ and of the hydrochloride salts of **5b** and **5c**.¹⁸ As depicted in Figure 2, use of imine **3** (R = Ph) resulted in the formation of a *syn*-adduct **5d**; whereas an *anti*-selectivity was observed when oxazolidines **2** (R = *i*-Pr or *t*-Bu) were used to afford amino alcohols **5b** and **5c**.

**Figure 2**

According to the general behavior of these organometallic species, the difference of regioselectivity observed between the Li and the Zn derivatives is not surprising.¹⁹ Indeed Li alkoxyallyl reagents are reported generally to react with carbonyl compounds to give mainly the γ -adducts when their Zn analogues usually react at the α -position (Scheme 3).^{2b}

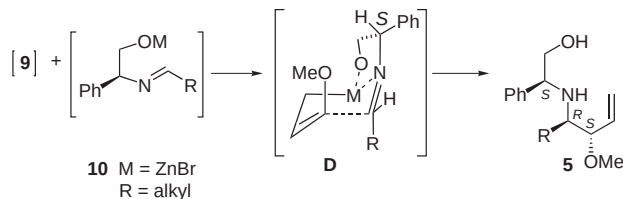
**Scheme 3**

The configuration of the newly created stereogenic center in the enol ether derivatives **4** can be assumed to be the result of a cyclic transition state **C** (Scheme 4) involving a metallic species **8** where the Li is located at the α -position relative to the OMe group. In such a transition state, due to chelation, this group is in axial position. Accordingly, the Li species, in a cisoid conformation, adds on the less hindered face of the *E*-imine **10**. The addition involves the participation of a chiral Li derivative, which is probably configurationally unstable.²⁰

**Scheme 4**

The stereoselection observed with the zinc derivative is more difficult to rationalize. In the cases where R is an alkyl group, it can be rationalized by a chair-like transition state, in which the steric interactions between the R and

the OMe groups are minimized. The attack, involving a metallic species **9** (Scheme 3) with the zinc located in γ -position and an enol ether in a Z-configuration occurs on the less hindered face of the *E*-imine, according to transition state **D** (Scheme 5). With R = Ph, different electronic and/or steric interactions must be taken into account. Mechanistic studies are currently underway to rationalize these contrasting results.



Scheme 5

In conclusion, by addition of Li or Zn reagents of methyl allyl ether to oxazolidines (or imines) derived from phenylglycinol, it is possible to prepare selectively the corresponding α - or γ -adducts with a good diastereoselectivity. Work is in progress to extend this methodology to the preparation of other substituted β -amino alcohols and to use them in the asymmetric synthesis of functionalized heterocycles.

Acknowledgment

We thank Dr. Boubekeur and Dr. Herson for X-ray analyses of compounds **4** (R = Ph), **5** (R = Ph) and **5** (R = *i*-Pr).

References and Notes

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- (11) **Typical Procedure for the Synthesis of Compounds 4.** *s*-BuLi (1.3 M in hexane-cyclohexane, 5.2 mL, 6.8 mmol) was added at -78°C to a solution of allyl methyl ether (0.60 mL, 6.4 mmol) in THF (10 mL). After stirring for 30 min at -78°C a solution of oxazolidine (2 mmol) in THF (10 mL) was added dropwise. After the reaction was complete, the mixture was quenched at -78°C by addition of sat. aq. NH_4Cl (15 mL). The aqueous layer was extracted with Et_2O (3×15 mL) and the organic layers were combined, dried over MgSO_4 and evaporated. The residue was then purified by chromatography on silica gel.
- (12) **Data for Compound 4d (R = Ph): [2S,2 (1S)]-2-(4-Methoxy-1-phenylbut-3-enylamino)-2-phenylethanol.** Solid, yield 67%; mp 63°C ; $[\alpha]_D^{20} +37$ (c 1.1, CHCl_3). ^1H NMR: δ = 7.33–7.20 (m, 10 H), 5.92 (dt, J = 6.3, 1.3 Hz, 1 H), 4.24 (dd, J = 7.3, 6.3 Hz, 1 H), 3.92 (dd, J = 7.1, 4.6 Hz, 1 H), 3.77 (dd, J = 10.6, 4.6 Hz, 1 H), 3.70 (t, J = 6.4 Hz, 1 H), 3.56 (s, 3 H), 3.53 (dd, J = 10.6, 7.3 Hz, 1 H), 2.9 (br d, 1 H), 2.62–2.45 (m, 2 H), 1.78 (ls, 1 H). ^{13}C NMR: 147.7, 144.0, 141.4, 128.5, 128.2, 127.4, 127.1, 127.0, 126.9, 102.5, 65.7, 61.3, 59.6, 59.5, 31.0. IR (CHCl_3): 3374, 3031, 2856, 1660, 1109 cm^{-1} . Anal. Calcd for $\text{C}_{19}\text{H}_{23}\text{NO}_2$: C, 76.73; H, 7.80; N, 4.71. Found: C, 76.60; H, 7.83; N, 4.69.

- (13) Atomic coordinates, bond lengths and angles, and thermal parameters have been deposited at the Cambridge Crystallographic Data Centre with the deposition number CCDC 293382.
- (14) This stereochemical outcome is well established for the attack of organometallic reagents onto phenylglycinol-derived oxazolidines, see, ref. 4b and 6a.
- (15) **Typical Procedure for the Synthesis of Compounds 5.** s-BuLi (1.3 M in hexane–cyclohexane, 5.2 mL, 6.8 mmol) was added at -78°C to a solution of allyl methyl ether (0.60 mL, 6.4 mmol) in THF (10 mL). After stirring for 30 min at -78°C , a solution of zinc bromide (1 M in THF, 7.2 mL, 7.2 mmol) was added. The mixture was stirred at -78°C for 40 min and then a solution of oxazolidine (2 mmol) in THF (10 mL) was added dropwise. After completion, the mixture was quenched (at -78°C for **5a** and **5d** and at r.t. for **5b** and **5c**) by addition of sat. aq NH_4Cl (20 mL). The aqueous layer was extracted with Et_2O (3×20 mL) and the organic layers were combined, dried over MgSO_4 and evaporated. The residue was purified by chromatography on silica gel.
- (16) Atomic coordinates, bond lengths and angles, and thermal parameters have been deposited at the Cambridge Crystallographic Data Centre with the deposition numbers CCDC 602015 for **5b**, CCDC 612649 for **5d** and CCDC 612650 for **5c**.
- (17) **Data for Compound 5d (R = Ph): [2S,2 (2S,1S)]-2-(2-Methoxy-1-phenylbut-3-enylamino)-2-phenylethanol.** Solid, yield 69%; mp 58°C ; $[\alpha]_{\text{D}}^{20} +38$ (c 1.1, CHCl_3). ^1H NMR: $\delta = 7.32\text{--}7.22$ (m, 10 H), 5.62–5.54 (m, 1 H), 5.28–5.20 (m, 2 H), 3.89–3.79 (m, 4 H), 3.61–3.56 (m, 1 H), 3.30 (s, 3 H). ^{13}C NMR: 141.6, 140.4, 135.5, 128.4, 128.3, 128.0, 127.2, 119.1, 86.1, 65.2, 63.8, 61.1, 56.8. IR (CHCl_3): 3390, 2925, 1764, 1602, 1453, 1094, 699 cm^{-1} . Anal. Calcd for $\text{C}_{19}\text{H}_{23}\text{NO}_2$: C, 76.73; H, 7.80; N, 4.71. Found: C, 76.54; H, 7.79; N, 4.54.
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